

Timolol Maleate Ophthalmic Solution 0.5% (UNIMS)

Description:

Clear, colourless solution. Filled in colourless LDPE unims.

Name of the active ingredients:

Each ml contains:

Timolol maleate USP 0.68% w/v equivalent to Timolol 0.5% w/v

Pharmacodynamics:

Timolol is a competitive antagonist at both beta 1 and beta 2 adrenoceptors: It is therefore a non-selective beta blocking drug. Timolol combines reversibly with beta-adrenergic receptor, and this inhibits the usual biologic response that would occur with stimulation of that receptor. This specific competitive antagonism blocks stimulation of the beta-adrenergic stimulating activity, whether these originate from an endogenous or exogenous source.

Pharmacokinetics:

The onset of reduction in intraocular pressure following administration of Timolol can usually be detected within one half hour after a single dose; maximum effect usually occurs after one to two hours.

Indication

Timolol Maleate Ophthalmic Solution is indicated in the treatment of elevated intraocular pressure in patients with ocular hypertension or open angle glaucoma.

Recommended dose

Instill one drop in the affected eye(s) twice daily or as directed by the physician.

Transfer from other agents

When another topical beta blocking agent is being used, discontinue its use after a full day of therapy and start treatment with Timolol Maleate Ophthalmic Solution the next day with one drop of 0.25% Timolol Maleate Ophthalmic Solution in each affected eye twice a day. The dosage may be increased to one drop of 0.5% solution in each affected eye twice a day if the response is not adequate.

When transferring a patient from a single anti glaucoma agent other than a topical beta blocking agent, continue the agent and add one drop of 0.25% Timolol Maleate

Ophthalmic Solution in each affected eye twice a day. On the following day, discontinue the previous agent completely, and continue with Iotim ED. If a higher dosage of Timolol Maleate Ophthalmic Solution is required, substitute one drop of 0.5% solution in each affected eye twice a day.

Route of administration

Ocular

Contraindication

Timolol is contraindicated in patients with bronchial asthma, or a history of bronchial asthma or severe chronic obstructive pulmonary disease, sinus bradycardia; second and third degree atrioventricular block; overt cardiac failure; cardiogenic shock; hypersensitivity to any component of the product.

Warning and precaution**(a) Precautions:**

Timolol Maleate Ophthalmic Solution should be used during pregnancy and lactation only if the potential benefit justifies the potential risk to the fetus. Because of the potential for serious adverse reactions from Timolol Maleate Ophthalmic Solution in nursing infants, a decision should be taken whether to discontinue nursing or to discontinue the drug taking into account the importance of the drug for the mother. Safety and effectiveness in pediatric patients have not been established.

(b) Warnings:

As with many topically applied ophthalmic drugs, this drug is absorbed systemically. The same adverse reactions found with systemic administration of beta-adrenergic blocking agents may occur with topical administration. For e.g. severe respiratory reactions and cardiac reactions including death due to bronchospasm in patients with asthma, and rarely death in association with cardiac failure have been reported following systemic or ophthalmic administration of Timolol.

Cardiac Failure: Sympathetic stimulation may be essential for support of the circulation in individuals with diminished myocardial contractility, and its inhibition by beta-adrenergic receptor blockade may precipitate more severe failure.

In patients without a history of cardiac failure continued depression of the myocardium with beta-blocking agents over a period of time, in some cases, can lead to cardiac failure.

Patients with chronic obstructive pulmonary disease (e.g. bronchitis, emphysema) of mild and moderate severity, bronchospastic disease, or history of bronchospastic disease in which Timolol is contraindicated should, in general, should not receive beta-blockers including Timolol Maleate Ophthalmic Solution.

Beta-adrenergic blocking agents should be administered with caution in patients subject to spontaneous hypoglycemia or to diabetic patients (especially those with labile diabetes).

Beta-adrenergic blocking agents may mask certain clinical signs of hyperthyroidism. Patients suspected of developing thyrotoxicosis should be managed carefully to avoid abrupt withdrawal of beta-adrenergic blocking agents that might precipitate a thyroid storm.

If irritation persists or increases, discontinue the use and consult the ophthalmologist.

Interaction with other medicaments

Although Timolol used alone has little or no effect on pupil size, mydriasis resulting from concomitant therapy with Timolol and epinephrine has been reported occasionally. Close observation of the patient is recommended when a beta blocker is administered to patients receiving catecholamine-depleting drugs such as reserpine, because of possible additive effects and the production of hypotension and/or marked bradycardia, which may produce vertigo, syncope or postural hypotension. The concomitant use of the beta-adrenergic blocking agents with digitalis and calcium antagonists may have additive effects in prolonging atrioventricular conduction time. Patients who are receiving beta adrenergic blocking agents orally and Timolol Maleate Ophthalmic Solution should be observed for potential additive effects of beta-blockade, both systemic and on intraocular pressure. The concomitant use of two topical beta adrenergic blocking agents is not recommended.

Pregnancy and lactation:

Timolol maleate has not been studied in human pregnancy. The use of Timolol maleate in pregnancy requires that the anticipated benefit be weighed against potential hazards.

Use in Lactation:

Timolol Maleate is detectable in human Milk. Because of the potential for serious adverse reactions in infants, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

Side effect

Local irritation is the only commonly observed adverse reaction. Bronchoconstriction is reported in elderly patients with pulmonary obstructive disease. **Timolol Maleate Ophthalmic Solution** is generally well tolerated following topical application to the eye. Signs and symptoms of ocular irritation, including burning & stinging sensation, conjunctivitis, blepharitis, keratitis and blepharotosis have been reported occasionally in patients receiving topical Timolol therapy. Visual disturbance including refractive changes (resulting from withdrawal of miotic therapy in some patients) have been infrequently associated with Timolol therapy. Decreased corneal sensitivity, diplopia and ptosis have also occurred. Aggravation or precipitation of certain cardiovascular pulmonary and other disorders, presumably related to effects of systemic beta-adrenergic blockade, may occur during therapy with topical Timolol and may include bradycardia, arrhythmia, congestive heart failure, hypotension, syncope, heart block, cerebrovascular accident, cerebral ischemia, palpitation, dyspnoea, nasal congestion, cough bronchospasm (mainly in patients with pre-existing bronchospastic disease) and respiratory failure. Headache, depression, dizziness, paresthesia, asthenia, diarrhoea and nausea have been reported in patients receiving topical Timolol therapy.

Symptoms for overdose

There have been reports of inadvertent overdosage of Timolol Maleate Ophthalmic drops resulting in systemic effects such as dizziness, headache, and shortness of breath, bradycardia, bronchospasm and cardiac arrest.

Overdose has been reported with timolol maleate tablets. A 30 year old female ingested 650 mg of Timolol maleate tablets (maximum recommended oral daily dose is 60 mg) had experienced second and third degree heart block, she recovered without treatment but approximately two months later developed irregular heartbeat, hypertension, dizziness, tinnitus, faintness, increased pulse rate and borderline first degree heart block. the oral LD50 of the drug is 1190 and 900 mg/kg in female mice and female rats respectively.

An invitro hemodialysis study, using timolol added to human plasma or whole blood, showed that timolol was readily dialyzed from these fluids; however, a study of patients with renal failure showed that timolol did not dialyze readily. The most common signs and symptoms to be expected with overdosage with administration of a systemic Beta adrenergic receptor blocking agent are symptomatic bradycardia, hypotension, bronchospasm and acute cardiac failure. The following therapeutic measures should be considered.

1. Gastric lavage: if ingested.
2. Symptomatic bradycardia: use atropine sulfate intravenously in a dosage of 0.25 mg to 2 mg. to induce vagal blockage. if bradycardia persists, intravenous isoproterenol hydrochloride should be administered cautiously. in refractory cases, the use of a transvenous cardiac pacemaker may be considered.
3. Hypotension: Use sympathomimetic pressor drug therapy such as dopamine, dobutamine or levarterenol. in refractory cases, the use of glucagon hydrochloride has been reported to be useful.
4. Bronchospasm: Use isoproterenol hydrochloride. Additional therapy with aminophylline may be considered.
5. Acute Cardiac Failure: Conventional therapy with digitalis, diuretics and oxygen should be instituted immediately. In refractory cases, the use of intravenous aminophylline is suggested. This may be followed if necessary by glucagon hydrochloride which has been reported to be useful.
6. Heart block (second or third degree): use isoproterenol hydrochloride or a transvenous cardiac pacemaker.

Storage condition

Do not store above 30°C.

Shelf life

24 months from the date of manufacture.

Presentation

5 x 6 x 0.4ml is packed in a printed carton along with a package insert.

Manufactured by :**FDC Limited**

B-8, M.I.D.C Industrial Area, Waluj,
Aurangabad- 431 136, Maharashtra, India

Product Registration Holder :

UNIMED SDN BHD

53, Jalan Tembaga SD 5/2B Bandar Sri Damansara,
52200, Kuala Lumpur

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